Lab – File System Commands

Introduction

In this lab, you will use CLI commands to manage files and folders in Windows.

Recommended Equipment

- A computer running Windows

Instructions

Step 1: Access the Windows command prompt.

a. Log on to a Windows computer. The account **ITEUser** is used as the example user account throughout this lab.

b. To access the Windows command prompt, click **Start** and enter **cmd**. Select **Command Prompt** or **cmd.exe** to continue.

Step 2: Create and change directories.

In this step, you will use the change directory (**cd**), make directory (**md**), and directory (**dir**) commands.

Note: A directory is another word for folder. Directory and folder are used interchangeably throughout this lab.

a. Type **cd** at the command prompt.

Question:

What is the current directory?

- *C:\Users\TIPQC*

```
C:\Users\TIPQC>cd
C:\Users\TIPQC
```

b. Type **dir** at the command prompt to list the files and folders that are in the current folder.

```
C:\Users\ITEUser> dir
 Volume in drive C has no label.
 Volume Serial Number is 9055-35E9

 Directory of C:\Users\ITEUser

04/27/2019  09:21 AM    <DIR>          .
04/27/2019  09:21 AM    <DIR>          ..
04/25/2019  11:39 AM    <DIR>          3D Objects
04/25/2019  11:39 AM    <DIR>          Contacts
04/26/2019  10:29 AM    <DIR>          Desktop
04/25/2019  11:39 AM    <DIR>          Documents
04/25/2019  11:39 AM    <DIR>          Downloads
<some output omitted>
               0 File(s)              0 bytes
```

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```
               15 Dir(s)  32,671,969,280 bytes free
```

```
 Directory of C:\Users\TIPQC

10/08/2026  11:00 AM    <DIR>          .
03/12/2026  01:42 PM    <DIR>          ..
01/23/2025  05:14 PM    <DIR>          .anaconda
03/03/2026  11:01 AM    <DIR>          .android
09/24/2025  10:45 AM    <DIR>          .cache
04/02/2025  02:49 PM    <DIR>          .conda
01/23/2025  05:14 PM    <DIR>          .continuum
07/25/2026  01:52 PM    <DIR>          .copilot
04/04/2025  02:03 PM    <DIR>          .ipynb_checkpoints
01/20/2025  11:45 AM    <DIR>          .ipython
02/07/2025  01:51 PM    <DIR>          .jupyter
02/07/2025  02:04 PM    <DIR>          .matplotlib
10/01/2025  04:06 PM               176 .packettracer
08/06/2025  05:25 PM    <DIR>          .spyder-py3
10/08/2026  11:03 AM    <DIR>          .VirtualBox
05/07/2025  02:22 PM    <DIR>          .vscode
07/18/2026  01:48 PM    <DIR>          .vscode-shared
10/01/2025  04:24 PM    <DIR>          Cisco Packet Tracer 8.2.2
10/18/2024  05:21 AM    <DIR>          Contacts
09/22/2026  08:54 PM    <DIR>          Desktop
10/03/2026  01:51 PM    <DIR>          Documents
10/08/2026  11:11 AM    <DIR>          Downloads
05/02/2025  08:28 AM    <DIR>          Favorites
02/10/2026  11:04 AM    <DIR>          flutter
10/08/2026  11:00 AM    <DIR>          GM
08/06/2025  04:53 PM    <DIR>          Jedi
10/18/2024  05:21 AM    <DIR>          Links
10/18/2024  05:21 AM    <DIR>          Music
10/18/2024  05:23 AM    <DIR>          OneDrive
05/26/2025  01:06 PM    <DIR>          Pictures
10/18/2024  05:21 AM    <DIR>          Saved Games
10/17/2024  03:58 PM    <DIR>          Searches
04/04/2025  03:00 PM    <DIR>          shapes
10/21/2024  01:20 PM    <DIR>          Videos
10/08/2026  11:03 AM    <DIR>          VirtualBox VMs
               1 File(s)            176 bytes
              34 Dir(s)  550,143,225,856 bytes free
```

c. In the current directory, use the **md** command to create three new folders: **ITEfolder1**, **ITEfolder2**, and **ITEfolder3**. Type **md ITEfolder1** and press **Enter**. Create **ITEfolder2** and **ITEfolder3**. (**Note**: The command **mkdir** performs the same function as the command **md**.)

```
C:\Users\ITEUser> md ITEfolder1
C:\Users\ITEUser> md ITEfolder2 ITEfolder3
```

```
C:\Users\TIPQC>md ITEfolder1

C:\Users\TIPQC>md ITEfolder2 ITEfolder3
```

d. Type **dir** to verify that the folders have been created.

```
10/08/2026  11:18 AM    <DIR>          ITEfolder1
10/08/2026  11:18 AM    <DIR>          ITEfolder2
10/08/2026  11:18 AM    <DIR>          ITEfolder3
```

e. Type **cd ITEfolder3** at the command prompt and press **Enter**.

Question:

Which folder are you in now?

- *I was directed to the "ITEfolder3" folder.*

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```
C:\Users\TIPQC>cd ITEfolder3

C:\Users\TIPQC\ITEfolder3>
```

f. Create **ITEfolder4** in the current directory. Within the **ITEfolder4** folder, create a folder named **ITEfolder5**. Use the **dir** command to verify the folder creation.

Question:

What command or commands did you use to create the nested folders?

- *I used the md command to create the ITEfolder4 inside the current folder which is the ITEfolder3, and then I used cd the cd command to get directed to ITEfolder4, and then I used the md command once again to create the ITEfolder5 inside the ITEfolder4.*

```
C:\Users\TIPQC\ITEfolder3>md ITEfolder4

C:\Users\TIPQC\ITEfolder3>cd ITEfolder4

C:\Users\TIPQC\ITEfolder3\ITEfolder4>md ITEfolder5
```

```
Directory of C:\Users\TIPQC\ITEfolder3\ITEfolder4

10/08/2026  11:21 AM    <DIR>          .
10/08/2026  11:21 AM    <DIR>          ..
10/08/2026  11:21 AM    <DIR>          ITEfolder5
               0 File(s)              0 bytes
               3 Dir(s)  550,300,483,584 bytes free
```

g. Change directory as necessary until you are in ITEfolder5.

```
C:\Users\TIPQC\ITEfolder3\ITEfolder4>cd ITEfolder5

C:\Users\TIPQC\ITEfolder3\ITEfolder4\ITEfolder5>
```

h. Type **cd ..** to change the current directory. Each **..** is a shortcut to move up one level in the directory tree.

Question:

After issuing the **cd ..** command, what is your directory now?

- *After using cd .., my current directory as of now is at ITEfolder4.*

```
C:\Users\TIPQC\ITEfolder3\ITEfolder4\ITEfolder5>cd ..

C:\Users\TIPQC\ITEfolder3\ITEfolder4>
```

What would be the current directory if you issue this command at C:\Users\ITEUser\ITEfolder3\ITEFolder4?

- *When I issue this command at ITEfolder4, I get directed back to ITEfolder3*

```
C:\Users\TIPQC\ITEfolder3\ITEfolder4>cd ..

C:\Users\TIPQC\ITEfolder3>
```

Step 3: Create text files.

a. Navigate to the **C:\Users\ITEUser\ITEfolder1** directory. Type **cd ..\ITEfolder1** at the prompt.

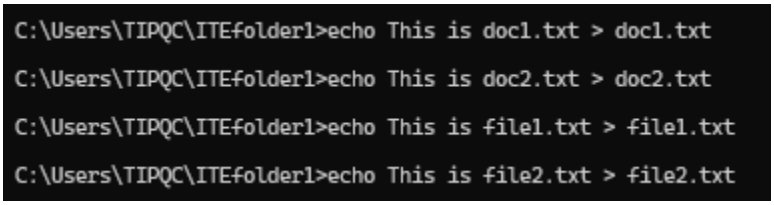
```
C:\Users\TIPQC\ITEfolder3>cd..\ITEfolder1

C:\Users\TIPQC\ITEfolder1>
```

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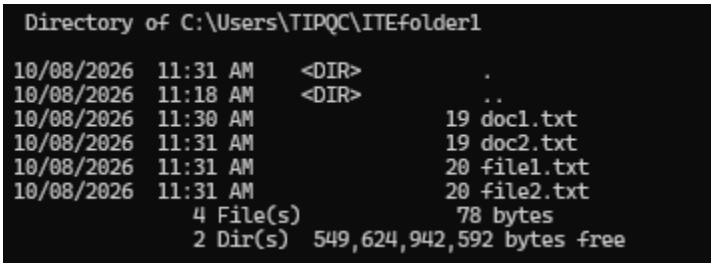
b. Type **echo This is doc1.txt > doc1.txt** at the command prompt. The **echo** command is used to display a message at the command prompt. The **>** is used to redirect the message from the screen to a file. For example, in the first line, the message **This is doc1.txt** is redirected into a new file named **doc1.txt**. Use the **echo** command and **>** redirect to create these files: **doc2.txt**, **file1.txt**, and **file2.txt**.

```
C:\Users\ITEUser\ITEfolder1> echo This is doc1.txt > doc1.txt
C:\Users\ITEUser\ITEfolder1> echo This is doc2.txt > doc2.txt
C:\Users\ITEUser\ITEfolder1> echo This is file1.txt > file1.txt
C:\Users\ITEUser\ITEfolder1> echo This is file2.txt > file2.txt
```



c. Use the **dir** command to verify the files are in the **ITEfolder1** folder.

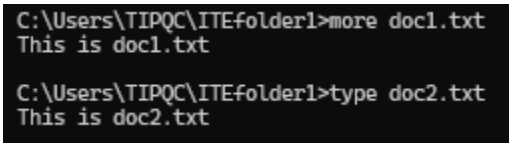
```
C:\Users\ITEUser\ITEFolder1> dir
<some output omitted>
04/29/2019  08:05 AM                19 doc1.txt
04/29/2019  08:06 AM                19 doc2.txt
04/29/2019  08:08 AM                20 file1.txt
04/29/2019  08:08 AM                20 file2.txt
               4 File(s)             78 bytes
               2 Dir(s)  32,625,397,760 bytes free
```



d. Use the **more** or **type** command to view the content of the newly created text files.

```
C:\Users\ITEUser\ITEfolder1> more doc1.txt
This is doc1.txt

C:\Users\ITEUser\ITEfolder1> type doc2.txt
This is doc2.txt
```



Step 4: Copy, delete, and move files.

a. At the command prompt, type **move doc2.txt C:\Users\ITEUser\ITEfolder2** to move the file **doc2.txt** to the **C:\Users\ITEUser\ITEfolder2** directory.

```
C:\Users\ITEUser\ITEfolder1> move doc2.txt C:\Users\ITEUser\ITEfolder2
```

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```
        1 file(s) moved.
```

```
C:\Users\TIPQC\ITEfolder1>move doc2.txt C:\Users\TIPQC\ITEfolder2
        1 file(s) moved.
```

b. Type **dir** at the prompt to verify that **doc2.txt** is no longer in the current directory.

```
 Directory of C:\Users\TIPQC\ITEfolder1

10/08/2026  11:35 AM    <DIR>          .
10/08/2026  11:18 AM    <DIR>          ..
10/08/2026  11:30 AM                19 doc1.txt
10/08/2026  11:31 AM                20 file1.txt
10/08/2026  11:31 AM                20 file2.txt
               3 File(s)             59 bytes
               2 Dir(s)  549,624,819,712 bytes free
```

c. Navigate to **C:\Users\ITEUser\ITEfolder2** to change the directory to **ITEfolder2**. Type **dir** at the prompt to verify **doc2.txt** has been moved.

```
 C:\Users\TIPQC\ITEfolder1>cd..\ITEfolder2

 C:\Users\TIPQC\ITEfolder2>dir
  Volume in drive C is Q3203-1
  Volume Serial Number is 16D7-17AB

  Directory of C:\Users\TIPQC\ITEfolder2

 10/08/2026  11:35 AM    <DIR>          .
 10/08/2026  11:18 AM    <DIR>          ..
 10/08/2026  11:31 AM                19 doc2.txt
                1 File(s)             19 bytes
                2 Dir(s)  549,624,680,448 bytes free
```

d. Type **copy doc2.txt doc2_copy.txt** to create a copy of **doc2.txt**. Type **dir** at the prompt to verify a copy of the file has been created.

```
C:\Users\ITEUser\ITEfolder2> dir
<some output omitted>
Directory of C:\Users\ITEUser\ITEfolder2

04/30/2019  09:07 AM                19 doc2.txt
04/30/2019  09:07 AM                19 doc2_copy.txt
               2 File(s)             38 bytes
               2 Dir(s)  31,753,068,544 bytes free
```

```
C:\Users\TIPQC\ITEfolder2>copy doc2.txt doc2_copy.txt
        1 file(s) copied.

C:\Users\TIPQC\ITEfolder2>dir
 Volume in drive C is Q3203-1
 Volume Serial Number is 16D7-17AB

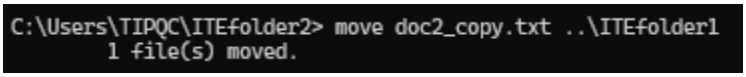
 Directory of C:\Users\TIPQC\ITEfolder2

10/08/2026  11:39 AM    <DIR>          .
10/08/2026  11:18 AM    <DIR>          ..
10/08/2026  11:31 AM                19 doc2.txt
10/08/2026  11:31 AM                19 doc2_copy.txt
               2 File(s)             38 bytes
               2 Dir(s)  549,620,232,192 bytes free
```

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e. Use the **move** command to move **doc2_copy.txt** to **ITEfolder1**. Type **move doc2_copy.txt ..\ITEfolder1**.

```
C:\Users\ITEUser\ITEfolder2> move doc2_copy.txt ..\ITEfolder1
        1 file(s) moved.
```



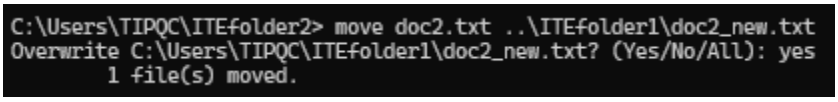
f. A copy of the file **doc2.txt** can be created in a new location using the **copy** command. At the prompt, enter the **copy doc2.txt ..\ITEfolder1\doc2_new.txt**.

```
C:\Users\ITEUser\ITEfolder2> copy doc2.txt ..\ITEfolder1\doc2_new.txt
        1 file(s) copied.
```



g. The file **doc2.txt** can also be moved to a new location with a new filename using the **move** command. Type **move doc2.txt ..\ITEfolder1\doc2_new.txt** at the prompt.

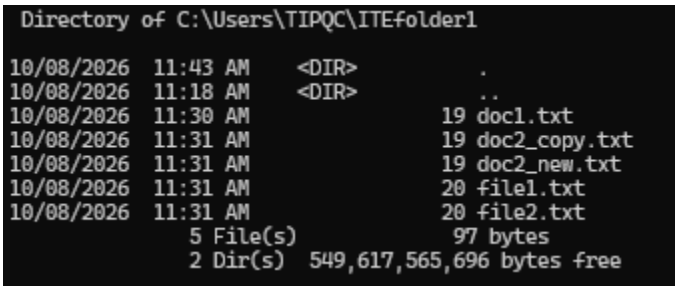
```
C:\Users\ITEUser\ITEfolder2> move doc2.txt ..\ITEfolder1\doc2_move.txt
        1 file(s) moved.
```



h. Type **dir ..\ITEfolder1** to view the content in **ITEfolder1** without leaving the current directory.

```
C:\Users\ITEUser\ITEfolder2> dir ..\ITEfolder1
<some output omitted>
 Directory of C:\Users\ITEUser\ITEfolder1

04/29/2019  12:08 PM    <DIR>          .
04/29/2019  12:08 PM    <DIR>          ..
04/29/2019  08:05 AM                19 doc1.txt
04/29/2019  08:06 AM                19 doc2_copy.txt
04/29/2019  08:06 AM                19 doc2_move.txt
04/29/2019  08:06 AM                19 doc2_new.txt
04/29/2019  08:08 AM                20 file1.txt
04/29/2019  08:08 AM                20 file2.txt
               6 File(s)            116 bytes
               2 Dir(s)  31,467,700,224 bytes free
```



i. Change the current directory to **ITEfolder1**. Type **cd ..\ITEfolder1** at the prompt.

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```
C:\Users\TIPQC\ITEfolder1>cd ..\ITEfolder1

C:\Users\TIPQC\ITEfolder1>
```

j. Move **file1.txt** and **file2.txt** into **ITEfolder3**. To move all the files that contain the word **file** into **ITEfolder3** with one command, use a **wildcard** (*) character to represent one or more characters. Type **move file* ..\ITEfolder3**.

```
C:\Users\ITEUser\ITEfolder1> move file* ..\ITEfolder3
C:\Users\ITEUser\ITEfolder1\file1.txt
C:\Users\ITEUser\ITEfolder1\file2.txt
        2 file(s) moved.
```

```
C:\Users\TIPQC\ITEfolder1>move file* ..\ITEfolder3.
C:\Users\TIPQC\ITEfolder1\file1.txt
C:\Users\TIPQC\ITEfolder1\file2.txt
        2 file(s) moved.
```

k. To view the content of the file doc2_new.txt, use the **type** or **more** command at the prompt.

```
C:\Users\ITEUser\ITEfolder1> type doc2_new.txt
This is doc2.txt
```

```
C:\Users\TIPQC\ITEfolder1>type doc2_new.txt
This is doc2.txt
```

l. Use the **ren** command to rename doc2_new.txt to doc3.txt. Use the **dir** command to display the content in the directory.

```
c:\Users\ITEUser\ITEfolder1> ren doc2_new.txt file.log
```

```
C:\Users\TIPQC\ITEfolder1>ren doc2_new.txt file.log

C:\Users\TIPQC\ITEfolder1>dir
 Volume in drive C is Q3203-1
 Volume Serial Number is 16D7-17AB

 Directory of C:\Users\TIPQC\ITEfolder1

10/08/2026  11:50 AM    <DIR>          .
10/08/2026  11:18 AM    <DIR>          ..
10/08/2026  11:30 AM                19 doc1.txt
10/08/2026  11:31 AM                19 doc2_copy.txt
10/08/2026  11:31 AM                19 file.log
               3 File(s)             57 bytes
               2 Dir(s)  549,615,104,000 bytes free
```

m. Use the **type** or **more** command to view the content of the newly rename file **file.log.**

```
C:\Users\ITEUser\ITEfolder1> more file.log
This is doc2.txt
```

```
C:\Users\TIPQC\ITEfolder1>more file.log
This is doc2.txt
```

n. To delete a file or multiple files, use the **del** command with the file names at the prompt.

Question:

What single command would you use to delete all the files with doc2 in the filename? Use the command to remove the files with doc2 in the filename.

We use del *doc2* to delete all the files with doc2 in the filename.

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```
C:\Users\TIPQC\ITEfolder1>del *doc2*

C:\Users\TIPQC\ITEfolder1>dir
 Volume in drive C is Q3203-1
 Volume Serial Number is 16D7-17AB

 Directory of C:\Users\TIPQC\ITEfolder1

10/08/2026  11:58 AM    <DIR>          .
10/08/2026  11:18 AM    <DIR>          ..
10/08/2026  11:30 AM                19 doc1.txt
10/08/2026  11:31 AM                19 file.log
               2 File(s)             38 bytes
               2 Dir(s)  549,612,556,288 bytes free
```

What command would you use to delete all the files in the directory? Use the command to remove the files.

- *To delete all the files in the directory, we use the command "del * ".*

```
C:\Users\TIPQC\ITEfolder1>del *
C:\Users\TIPQC\ITEfolder1\*, Are you sure (Y/N)? Y

C:\Users\TIPQC\ITEfolder1>dir
 Volume in drive C is Q3203-1
 Volume Serial Number is 16D7-17AB

 Directory of C:\Users\TIPQC\ITEfolder1

10/08/2026  11:59 AM    <DIR>          .
10/08/2026  11:18 AM    <DIR>          ..
               0 File(s)              0 bytes
               2 Dir(s)  549,613,015,040 bytes free
```

Step 5: Use the xcopy and robocopy commands.

In this step, the **xcopy** and **robocopy** commands are used to copy the content in a directory.

a. View the content of ITEfolder3.

```
C:\Users\ITEUser\ITEfolder1> dir ..\ITEfolder3
<some output omitted>
 Directory of c:\Users\ITEUser\ITEfolder3

04/29/2019  03:11 PM    <DIR>          .
04/29/2019  03:11 PM    <DIR>          ..
04/29/2019  03:05 PM                20 file1.txt
04/29/2019  03:05 PM                20 file2.txt
04/29/2019  03:01 PM    <DIR>          ITEfolder4
               2 File(s)             40 bytes
               3 Dir(s)  31,492,157,440 bytes free
```

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```
C:\Users\TIPQC\ITEfolder1>dir ..\ITEfolder3
 Volume in drive C is Q3203-1
 Volume Serial Number is 16D7-17AB

 Directory of C:\Users\TIPQC\ITEfolder3

10/08/2026  11:47 AM    <DIR>          .
10/08/2026  11:18 AM    <DIR>          ..
10/08/2026  11:31 AM                20 file1.txt
10/08/2026  11:31 AM                20 file2.txt
10/08/2026  11:21 AM    <DIR>          ITEfolder4
               2 File(s)             40 bytes
               3 Dir(s)  549,398,278,144 bytes free
```

b. Type **xcopy ..\ITEfolder3 .** at the prompt to copy the content of **ITEfolder3** to **ITEfolder1**. Note the **.** at the end of the command. It is a shortcut for the current directory.

```
C:\Users\ITEUser\ITEfolder1> xcopy ..\ITEfolder3 .
..\ITEfolder3\file1.txt
..\ITEfolder3\file2.txt
2 File(s) copied
```

```
C:\Users\TIPQC\ITEfolder1>xcopy ..\ITEfolder3
..\ITEfolder3\file1.txt
..\ITEfolder3\file2.txt
2 File(s) copied
```

c. At the prompt, type **dir** to display the content of **ITEfolder1**. Only the files in the **ITEfolder3** were copied into **ITEfolder1**. The directory **ITEfolder4** was not copied into **ITEfolder3**.

```
C:\Users\ITEUser\ITEfolder1> dir
<some output omitted>
 Directory of C:\Users\ITEUser\ITEfolder1
04/29/2019  03:16 PM    <DIR>          .
04/29/2019  03:16 PM    <DIR>          ..
04/29/2019  03:05 PM                20 file1.txt
04/29/2019  03:05 PM                20 file2.txt
               2 File(s)             40 bytes
               2 Dir(s)  31,491,321,856 bytes free
```

```
C:\Users\TIPQC\ITEfolder1>dir
 Volume in drive C is Q3203-1
 Volume Serial Number is 16D7-17AB

 Directory of C:\Users\TIPQC\ITEfolder1

10/08/2026  12:01 PM    <DIR>          .
10/08/2026  11:18 AM    <DIR>          ..
10/08/2026  11:31 AM                20 file1.txt
10/08/2026  11:31 AM                20 file2.txt
               2 File(s)             40 bytes
               2 Dir(s)  549,397,200,896 bytes free
```

d. Use **help xcopy** or **xcopy /?** to determine which switch would allow the **xcopy** command to copy **all** the files and directories.

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```
C:\Users\TIPQC\ITEfolder1>help xcopy
Copies files and directory trees.

XCOPY source [destination] [/A | /M] [/D[:date]] [/P] [/S [/E]] [/V] [/W]
                           [/C] [/I] [/-I] [/Q] [/F] [/L] [/G] [/H] [/R] [/T]
                           [/U] [/K] [/N] [/O] [/X] [/Y] [/-Y] [/Z] [/B] [/J]
                           [/EXCLUDE:file1[+file2][+file3]...] [/COMPRESS]
                           [/[-]SPARSE] [/NOCLONE]

  source       Specifies the file(s) to copy.
  destination  Specifies the location and/or name of new files.
  /A           Copies only files with the archive attribute set,
               doesn't change the attribute.
  /M           Copies only files with the archive attribute set,
               turns off the archive attribute.
  /D:m-d-y     Copies files changed on or after the specified date.
               If no date is given, copies only those files whose
               source time is newer than the destination time.
  /EXCLUDE:file1[+file2][+file3]...
               Specifies a list of files containing strings.  Each string
               should be in a separate line in the files.  When any of the
               strings match any part of the absolute path of the file to be
               copied, that file will be excluded from being copied.  For
               example, specifying a string like \obj\ or .obj will exclude
               all files underneath the directory obj or all files with the
               .obj extension respectively.
  /P           Prompts you before creating each destination file.
  /S           Copies directories and subdirectories except empty ones.
  /E           Copies directories and subdirectories, including empty ones.
               Same as /S /E. May be used to modify /T.
  /V           Verifies the size of each new file.
  /W           Prompts you to press a key before copying.
  /C           Continues copying even if errors occur.
  /I           If destination does not exist and copying more than one file,
               assumes that destination must be a directory.
  /-I          If destination does not exist and copying a single specified file,
               assumes that destination must be a file.
  /Q           Does not display file names while copying.
  /F           Displays full source and destination file names while copying.
  /L           Displays files that would be copied.
  /G           Allows the copying of encrypted files to destination that does
               not support encryption.
  /H           Copies hidden and system files also.
  /R           Overwrites read-only files.
  /T           Creates directory structure, but does not copy files. Does not
               include empty directories or subdirectories. /T /E includes
               empty directories and subdirectories.
  /U           Copies only files that already exist in destination.
  /K           Copies attributes. Normal Xcopy will reset read-only attributes.
  /N           Copies using the generated short names.
  /O           Copies file ownership and ACL information.
  /X           Copies file audit settings (implies /O).
  /Y           Suppresses prompting to confirm you want to overwrite an
               existing destination file.
  /-Y          Causes prompting to confirm you want to overwrite an
               existing destination file.
  /Z           Copies networked files in restartable mode.
  /B           Copies the Symbolic Link itself versus the target of the link.
  /J           Copies using unbuffered I/O. Recommended for very large files.
  /COMPRESS    Request network compression during file transfer where
               applicable.
  /[-]SPARSE   Enable or disable retaining the sparse state of files during
               copy. /-SPARSE overrides /SPARSE if both are provided.
  /NOCLONE     Do not attempt block cloning as an optimization.

The switch /Y may be preset in the COPYCMD environment variable.
This may be overridden with /-Y on the command line.
```

Question:

What option allows you to copy all the files and directories, including the empty directories?

- *The /E option allows us to copy directories and subdirectories, including empty ones.*

```
/E           Copies directories and subdirectories, including empty ones.
             Same as /S /E. May be used to modify /T.
```

e. Because **ITEfolder4** is a subfolder and **ITEfolder5** is both a subfolder and an empty folder, **/E** is needed to copy all the contents of **ITEfolder3** and the empty subfolder.

Type **xcopy /E ..\ITEfolder3 .** at the prompt to copy the files. When prompted, type **a** to allow overwriting the existing files.

```
c:\Users\ITEUser\ITEfolder1> xcopy /E ..\ITEfolder3
Overwrite C:\Users\ITEUser\ITEfolder1\file1.txt (Yes/No/All)? a
..\ITEfolder3\file1.txt
```

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```
..\ITEfolder3\file2.txt
2 File(s) copied
```

```
C:\Users\TIPQC\ITEfolder1>xcopy /E ..\ITEfolder3
Overwrite C:\Users\TIPQC\ITEfolder1\file1.txt (Yes/No/All)? a
..\ITEfolder3\file1.txt
..\ITEfolder3\file2.txt
2 File(s) copied
```

f. Verify that the **ITEfolder4** and **ITEfolder5** were also copied in **ITEfolder1**.

```
c:\Users\ITEUser\ITEfolder1> dir
<some output omitted>
 Directory of c:\Users\ITEUser\ITEfolder1

04/29/2019  04:41 PM    <DIR>          .
04/29/2019  04:41 PM    <DIR>          ..
04/29/2019  03:05 PM                20 file1.txt
04/29/2019  03:05 PM                20 file2.txt
04/29/2019  04:41 PM    <DIR>          ITEfolder4
               2 File(s)             40 bytes
               3 Dir(s)  31,493,193,728 bytes free

c:\Users\ITEUser\ITEfolder1> dir ITEfolder4
<some output omitted>
 Directory of c:\Users\ITEUser\ITEfolder1\ITEfolder4

04/29/2019  04:41 PM    <DIR>          .
04/29/2019  04:41 PM    <DIR>          ..
04/29/2019  03:00 PM    <DIR>          ITEfolder5
               0 File(s)              0 bytes
               3 Dir(s)  31,493,193,728 bytes free
```

```
C:\Users\TIPQC\ITEfolder1>dir
 Volume in drive C is Q3203-1
 Volume Serial Number is 16D7-17AB

 Directory of C:\Users\TIPQC\ITEfolder1

10/08/2026  12:06 PM    <DIR>          .
10/08/2026  11:18 AM    <DIR>          ..
10/08/2026  11:31 AM                20 file1.txt
10/08/2026  11:31 AM                20 file2.txt
10/08/2026  12:06 PM    <DIR>          ITEfolder4
               2 File(s)             40 bytes
               3 Dir(s)  549,399,891,968 bytes free

C:\Users\TIPQC\ITEfolder1>dir ITEfolder4
 Volume in drive C is Q3203-1
 Volume Serial Number is 16D7-17AB

 Directory of C:\Users\TIPQC\ITEfolder1\ITEfolder4

10/08/2026  12:06 PM    <DIR>          .
10/08/2026  12:06 PM    <DIR>          ..
10/08/2026  11:21 AM    <DIR>          ITEfolder5
               0 File(s)              0 bytes
               3 Dir(s)  549,399,773,184 bytes free
```

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g. The robocopy command can also be used to copy the content of a directory to a new destination. The robocopy command has more capabilities than the built-in Windows copy and xcopy command, such as resume copying after a network interruption, skip files that appear to be identical to the files in the destination folders, and mirror a directory by keeping the destination directory in sync with the source directory.

Copy the ITEfolder4 content to ITEUser folder using the robocopy command.

```
C:\Users\ITEUser\ITEfolder1> robocopy /E
C:\Users\ITEUser\ITEfolder3\ITEfolder4\ C:\Users\ITEUser
```

Note the information provided by the command during the copying process.



```
C:\Users\TIPQC\ITEfolder1>robocopy /E C:\Users\TIPQC\ITEfolder3\ITEfolder4\ C:\Users\TIPQC

------------------------------------------------------------------------------
   ROBOCOPY     ::     Robust File Copy for Windows
------------------------------------------------------------------------------

  Started : Thursday, October 8, 2026 12:09:43 PM
   Source : C:\Users\TIPQC\ITEfolder3\ITEfolder4\
     Dest : C:\Users\TIPQC\

    Files : *.*

  Options : *.* /S /E /DCOPY:DA /COPY:DAT /R:1000000 /W:30

------------------------------------------------------------------------------

                    0    C:\Users\TIPQC\ITEfolder3\ITEfolder4\
        *EXTRA Dir        -1    C:\Users\TIPQC\.anaconda\
        *EXTRA Dir        -1    C:\Users\TIPQC\.android\
        *EXTRA Dir        -1    C:\Users\TIPQC\.cache\
        *EXTRA Dir        -1    C:\Users\TIPQC\.conda\
        *EXTRA Dir        -1    C:\Users\TIPQC\.continuum\
        *EXTRA Dir        -1    C:\Users\TIPQC\.copilot\
        *EXTRA Dir        -1    C:\Users\TIPQC\.ipynb_checkpoints\
        *EXTRA Dir        -1    C:\Users\TIPQC\.ipython\
        *EXTRA Dir        -1    C:\Users\TIPQC\.jupyter\
        *EXTRA Dir        -1    C:\Users\TIPQC\.matplotlib\
        *EXTRA Dir        -1    C:\Users\TIPQC\.spyder-py3\
        *EXTRA Dir        -1    C:\Users\TIPQC\.VirtualBox\
        *EXTRA Dir        -1    C:\Users\TIPQC\.vscode\
        *EXTRA Dir        -1    C:\Users\TIPQC\.vscode-shared\
        *EXTRA Dir        -1    C:\Users\TIPQC\AppData\
        *EXTRA Dir        -1    C:\Users\TIPQC\Application Data\
        *EXTRA Dir        -1    C:\Users\TIPQC\Cisco Packet Tracer 8.2.2\
        *EXTRA Dir        -1    C:\Users\TIPQC\Contacts\
        *EXTRA Dir        -1    C:\Users\TIPQC\Cookies\
        *EXTRA Dir        -1    C:\Users\TIPQC\Desktop\
        *EXTRA Dir        -1    C:\Users\TIPQC\Documents\
        *EXTRA Dir        -1    C:\Users\TIPQC\Downloads\
        *EXTRA Dir        -1    C:\Users\TIPQC\Favorites\
        *EXTRA Dir        -1    C:\Users\TIPQC\flutter\
        *EXTRA Dir        -1    C:\Users\TIPQC\GM\
        *EXTRA Dir        -1    C:\Users\TIPQC\ITEfolder1\
        *EXTRA Dir        -1    C:\Users\TIPQC\ITEfolder2\
        *EXTRA Dir        -1    C:\Users\TIPQC\ITEfolder3\
        *EXTRA Dir        -1    C:\Users\TIPQC\Jedi\
        *EXTRA Dir        -1    C:\Users\TIPQC\Links\
        *EXTRA Dir        -1    C:\Users\TIPQC\Local Settings\
        *EXTRA Dir        -1    C:\Users\TIPQC\Music\
        *EXTRA Dir        -1    C:\Users\TIPQC\My Documents\
        *EXTRA Dir        -1    C:\Users\TIPQC\NetHood\
        *EXTRA Dir        -1    C:\Users\TIPQC\OneDrive\
        *EXTRA Dir        -1    C:\Users\TIPQC\Pictures\
        *EXTRA Dir        -1    C:\Users\TIPQC\PrintHood\
        *EXTRA Dir        -1    C:\Users\TIPQC\Recent\
        *EXTRA Dir        -1    C:\Users\TIPQC\Saved Games\
        *EXTRA Dir        -1    C:\Users\TIPQC\Searches\
        *EXTRA Dir        -1    C:\Users\TIPQC\SendTo\
        *EXTRA Dir        -1    C:\Users\TIPQC\shapes\
        *EXTRA Dir        -1    C:\Users\TIPQC\Start Menu\
        *EXTRA Dir        -1    C:\Users\TIPQC\Templates\
        *EXTRA Dir        -1    C:\Users\TIPQC\Videos\
        *EXTRA Dir        -1    C:\Users\TIPQC\VirtualBox VMs\
          *EXTRA File             176        .packettracer
          *EXTRA File            17.5 m      NTUSER.DAT
          *EXTRA File             4.3 m      ntuser.dat.LOG1
          *EXTRA File             4.4 m      ntuser.dat.LOG2
          *EXTRA File            73728       NTUSER.DAT{f6193da3-64f4-11ef-b9a5-d88e791e1f2b}.TM.blf
          *EXTRA File             4096       NTUSER.DAT{f6193da3-64f4-11ef-b9a5-d88e791e1f2b}.TM.blf.cnpf
          *EXTRA File           561152       NTUSER.DAT{f6193da3-64f4-11ef-b9a5-d88e791e1f2b}.TMContainer00000000000000000001.regtrans-ms
          *EXTRA File             4096       NTUSER.DAT{f6193da3-64f4-11ef-b9a5-d88e791e1f2b}.TMContainer00000000000000000001.regtrans-ms.cnpf
          *EXTRA File           561152       NTUSER.DAT{f6193da3-64f4-11ef-b9a5-d88e791e1f2b}.TMContainer00000000000000000002.regtrans-ms
          *EXTRA File             4096       NTUSER.DAT{f6193da3-64f4-11ef-b9a5-d88e791e1f2b}.TMContainer00000000000000000002.regtrans-ms.cnpf
          *EXTRA File               20       ntuser.ini
          New Dir          0    C:\Users\TIPQC\ITEfolder3\ITEfolder4\ITEfolder5\

------------------------------------------------------------------------------

               Total    Copied   Skipped  Mismatch    FAILED    Extras
    Dirs :         2         1         1         0         0        46
   Files :         0         0         0         0         0        11
   Bytes :         0         0         0         0         0   27.44 m
   Times :   0:00:00   0:00:00                       0:00:00   0:00:00
   Ended : Thursday, October 8, 2026 12:09:43 PM
```

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h. Navigate to **C:\Users\ITEUser** to verify that the folder ITEfolder5 is copied.

```
C:\Users\TIPQC\ITEfolder1>cd ..

C:\Users\TIPQC>dir
 Volume in drive C is Q3203-1
 Volume Serial Number is 16D7-17AB

 Directory of C:\Users\TIPQC

10/08/2026  12:09 PM    <DIR>          .
03/12/2026  01:42 PM    <DIR>          ..
01/23/2025  05:14 PM    <DIR>          .anaconda
03/03/2026  11:01 AM    <DIR>          .android
09/24/2025  10:45 AM    <DIR>          .cache
04/02/2025  02:49 PM    <DIR>          .conda
01/23/2025  05:14 PM    <DIR>          .continuum
07/25/2026  01:52 PM    <DIR>          .copilot
04/04/2025  02:03 PM    <DIR>          .ipynb_checkpoints
01/20/2025  11:45 AM    <DIR>          .ipython
02/07/2025  01:51 PM    <DIR>          .jupyter
02/07/2025  02:04 PM    <DIR>          .matplotlib
10/01/2025  04:06 PM               176 .packettracer
08/06/2025  05:25 PM    <DIR>          .spyder-py3
10/08/2026  11:03 AM    <DIR>          .VirtualBox
05/07/2025  02:22 PM    <DIR>          .vscode
07/18/2026  01:48 PM    <DIR>          .vscode-shared
10/01/2025  04:24 PM    <DIR>          Cisco Packet Tracer 8.2.2
10/18/2024  05:21 AM    <DIR>          Contacts
09/22/2026  08:54 PM    <DIR>          Desktop
10/03/2026  01:51 PM    <DIR>          Documents
10/08/2026  11:11 AM    <DIR>          Downloads
05/02/2025  08:28 AM    <DIR>          Favorites
02/10/2026  11:04 AM    <DIR>          flutter
10/08/2026  11:00 AM    <DIR>          GM
10/08/2026  12:06 PM    <DIR>          ITEfolder1
10/08/2026  11:43 AM    <DIR>          ITEfolder2
10/08/2026  11:47 AM    <DIR>          ITEfolder3
10/08/2026  11:21 AM    <DIR>          ITEfolder5
08/06/2025  04:53 PM    <DIR>          Jedi
10/18/2024  05:21 AM    <DIR>          Links
10/18/2024  05:21 AM    <DIR>          Music
10/18/2024  05:23 AM    <DIR>          OneDrive
05/26/2025  01:06 PM    <DIR>          Pictures
10/18/2024  05:21 AM    <DIR>          Saved Games
10/17/2024  03:58 PM    <DIR>          Searches
04/04/2025  03:00 PM    <DIR>          shapes
10/21/2024  01:20 PM    <DIR>          Videos
10/08/2026  11:03 AM    <DIR>          VirtualBox VMs
               1 File(s)            176 bytes
              38 Dir(s)  549,393,195,008 bytes free
```

Step 6: Delete directories.

In this step, you will delete an empty and a non-empty directory using the **rd** command.

a. Use the **rd ITEfolder2** to delete the empty directory and verify that the directory was deleted.

```
C:\Users\ITEUser> rd ITEfolder2
```

```
C:\Users\TIPQC>rd ITEfolder2

C:\Users\TIPQC>
```

b. Navigate to the **C:\Users\ITEUser\ITEfolder3** directory.

```
C:\Users\TIPQC>cd ITEfolder3

C:\Users\TIPQC\ITEfolder3>
```

c. Use the **rd ITEfolder1** to delete the directory. Verify the directory removal using the **dir** command.

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```
C:\Users\TIPQC\ITEfolder3>rd ITEfolder1
The system cannot find the file specified.
```

Were you able to delete the directory? Explain.

- *No, as the system cannot find the specified file.*

d. Use **rd /?** command to determine the switch that allows the deletion of a non-empty directory.

```
c:\Users\ITEUser\ITEfolder3> rd /S ITEfolder1
ITEfolder4, Are you sure (Y/N)? Y
```

```
C:\Users\TIPQC\ITEfolder3>rd /S ITEfolder1
ITEfolder1, Are you sure (Y/N)? Y
The system cannot find the file specified.
```

e. Use the appropriate commands to delete all the text files and folders that you have created in this lab. Type **exit** to close the command prompt window.

```
C:\Users\TIPQC>rmdir /S ITEfolder1
ITEfolder1, Are you sure (Y/N)? y

C:\Users\TIPQC>cd ITEfolder1
The system cannot find the path specified.

C:\Users\TIPQC>rmdrir /S ITEFolder2
'rmdrir' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\TIPQC>rmdir /S ITEfolder2
ITEfolder2, Are you sure (Y/N)? y
The system cannot find the file specified.

C:\Users\TIPQC>rmdir /S ITEfolder3
ITEfolder3, Are you sure (Y/N)? y

C:\Users\TIPQC>rmdir /S ITEfolder4
ITEfolder4, Are you sure (Y/N)? y
The system cannot find the file specified.

C:\Users\TIPQC>rmdir /S ITEfolder5
ITEfolder5, Are you sure (Y/N)? y

C:\Users\TIPQC>exit
```

Reflection Question

What are the advantages of using CLI vs. GUI?

- *Using CLI has several advantages over GUI. CLI can be faster and more efficient because users can perform tasks by typing commands directly instead of navigating through different menus, folders, or buttons. It allows users to have more control over the computer. However, GUI is generally easier for beginners because it is more visual and user-friendly. Overall, CLI is useful for users who want speed, control, and efficiency, while GUI is better for an easier user experience.*

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